A two-scale scheme for the Cox Boolean model on Poisson lines

Working note

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Abstract

We adapt Gouéré's two-scale argument for the subcritical phase of the Poisson Boolean model to the Boolean model driven by a Cox process on a Poisson line process. The argument is run directly on occupied crossings of annuli, so that neither thick clusters, nor the skeleton bound, nor the first moment over cluster chains of our earlier notes is needed. Two things change relative to the Poisson Boolean model. The inner and outer regions of the original argument are replaced by two bounded regions of the same size, because every line through a bounded region reaches its complement; and the dependence between the two regions, which in the original argument is carried by rare large balls, is carried here by the lines meeting both regions, a constant fraction of all lines through either, and is handled by conditioning on them and treating them as fixed lines. The induction hypothesis therefore carries fixed lines with a weight, and we show that the recursion then becomes a deterministic inequality for the pair (target function, weight) in which randomness enters only through the law of the shared lines. Defining the weight recursively makes the crossing bound unconditional and reduces the existence of a subcritical phase to a single statement, that the recursively defined target function tends to zero. We prove the inclusion lemma for any constant number a of regions, the decoupling, the bounds on shared lines, the recursion theorem, and the closure of one recursion step under a group-weight hypothesis on the shared lines; we show that the pointwise group-weight hypothesis cannot propagate because of configurations that make a sub-region free, that for $a \geq 3$ such configurations are rare enough to be discarded in expectation, and we formulate the inductive statement, a group-weight bound in expectation, whose propagation is equivalent to the success of the scheme. We also correct a claim of our multiscale note: a polynomial or faster target does not force many sub-boxes at polynomially separated scales; a constant number of regions at a fixed ratio suffices, because the product term of the recursion is far below the target.

1 Setting

Let $\mathcal{L}$ be a Poisson line process in $\mathbb{R}^2$ with intensity measure $\nu(\,\mathrm{d}\theta\,\mathrm{d}p) = (\mu/\pi)\,\mathrm{d}\theta\,\mathrm{d}p$, so that the lines meeting a convex set of perimeter P form a Poisson variable with mean $\mu P/\pi$; the lines

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meeting a disk of radius r have measure $2\mu r$. For a finite set G of lines (*fixed lines*), let X be, conditionally on $\mathcal{L}$, a Poisson process of intensity λ per unit length on every line of $\mathcal{L} \cup G$. For a Borel set A let

$$\Xi(A) \coloneqq \bigcup_{c \in X \cap A} B(c,1)$$

be the union of the unit disks centred at the points of X in A, and $\Xi \coloneqq \Xi(\mathbb{R}^2)$. We write $\mathbb{P}_G$ for the law of the model with fixed lines G and $\mathbb{P} = \mathbb{P}_\varnothing$. The radius 1 is no restriction by scaling.

Crossing events. Following [Gou08], for $x \in \mathbb{R}^2$ and $\alpha \geq 1$ let $\mathcal{E}(x,\alpha)$ be the event that the connected component of

$$\Xi\big(B(x,10\alpha)\big) \cup B(x,\alpha)$$

containing x is not contained in $B(x,8\alpha)$. The event depends only on the lines of $\mathcal{L} \cup G$ meeting $B(x,10\alpha)$ and on the points of X in $B(x,10\alpha)$. Put

$$\pi_\alpha(x;G) \coloneqq \mathbb{P}_G(\mathcal{E}(x,\alpha))\,, \qquad \pi_\alpha \coloneqq \pi_\alpha(0;\varnothing)\,. \tag{1.1}$$

Without fixed lines the model is stationary and $\pi_\alpha(x;\varnothing) = \pi_\alpha$.

Lemma 1.1 (Percolation forces crossings)**.** *Let $\alpha \geq 1$. If the component of Ξ containing a point of $B(0,1)$ is unbounded, then $\mathcal{E}(0,\alpha)$ occurs. Consequently, if $\pi_\alpha \to 0$ as $\alpha \to \infty$, then Ξ has no unbounded component almost surely.*

Proof. Let $D_1, D_2, \dots$ be a chain of pairwise consecutively intersecting disks of Ξ with $D_1 \cap B(0,1) \neq \varnothing$ that leaves $B(0,9\alpha)$; such a chain exists if the component is unbounded, since the component is a union of disks and is path connected through consecutive intersections. Let D_n be the first disk of the chain not contained in $B(0,8\alpha)$. The centres of $D_1, \dots, D_{n-1}$ lie in $B(0,8\alpha)$, and the centre of D_n is within distance 1 of a point of $B(0,8\alpha)$, since D_n meets D_{n-1} or, if $n = 1$, meets $B(0,1)$; so all these centres lie in $B(0,10\alpha)$ for $\alpha \geq 1$. The set $B(0,\alpha) \cup D_1 \cup \cdots \cup D_n$ is connected, contains 0 and is not contained in $B(0,8\alpha)$, which is $\mathcal{E}(0,\alpha)$. For the consequence, if an unbounded component existed with positive probability, by stationarity it would meet $B(z,1)$ for some $z \in \mathbb{Z}^2$ with positive probability, hence $B(0,1)$ with positive probability, and then $\pi_\alpha \geq \mathbb{P}(\text{unbounded component meets } B(0,1)) > 0$ for every α. □

2 The multi-region inclusion

Fix an integer $a \geq 2$ and put

$$M \coloneqq 10(a-1), \qquad r_i \coloneqq M + 70(i-1) \quad (i = 1, \dots, a)\,, \tag{2.1}$$

so that $r_1 = M$ and $r_a = 8M$; for $a = 2$ this is Gouéré's choice $M = 10$, $r_1 = 10$, $r_2 = 80$. The number M is the ratio between consecutive scales of the recursion. For each i fix a finite set $K_i \subseteq S_{r_i}$ of points on the circle of radius r_i around the origin with $S_{r_i} \subseteq K_i + B(0,1)$; one can take $|K_i| \leq 7r_i + 1$. Put $\mathcal{K} \coloneqq K_1 \times \cdots \times K_a$ and $N_\mathcal{K} \coloneqq \prod_i |K_i|$. For $x \in \mathbb{R}^2$, $\alpha \geq 1$ and $k = (k_1, \dots, k_a) \in \mathcal{K}$ write

$$x_i \coloneqq x + \alpha k_i, \qquad R_i \coloneqq B(x_i, 10\alpha) \qquad (i = 1, \dots, a)\,.$$

Two points on circles of radii $r_i\alpha$ and $r_j\alpha$ are at distance at least $|r_i - r_j|\alpha$, so $|x_i - x_j| \geq 70\alpha$ for $i \neq j$, the regions R_i are pairwise at distance at least 50α, and all of them lie in $B(x, 10M\alpha)$ since $8M + 10 \leq 10M$.

Lemma 2.1 (Inclusion, after [Gou08, Lemma 3.3])**.** *For every finite set G of lines, every $x \in \mathbb{R}^2$ and $\alpha \geq 1$,*

$$\mathcal{E}(x, M\alpha) \subseteq \bigcup_{k \in \mathcal{K}} \bigcap_{i=1}^{a} \mathcal{E}(x_i, \alpha)\,. \tag{2.2}$$

In [Gou08] the inclusion holds only outside the event that a ball of radius at least α is present, and that event supplies the error term of the recursion. Here all disks have radius $1 \leq \alpha$ and the inclusion holds outright.

Proof. Assume $\mathcal{E}(x, M\alpha)$ and take $x = 0$. The component of $\Xi(B(0, 10M\alpha)) \cup B(0, M\alpha)$ containing 0 is not contained in $B(0, 8M\alpha)$; since $X \cap B(0, 10M\alpha)$ is finite almost surely, there is a chain $D_1, \dots, D_n$ of unit disks with centres in $B(0, 10M\alpha)$, consecutive disks intersecting, with $D_1 \cap B(0, M\alpha) \neq \varnothing$ and $D_n \not\subseteq B(0, 8M\alpha)$. The union $U := D_1 \cup \cdots \cup D_n$ is connected and contains a point of $B(0, M\alpha)$ and a point outside $B(0, 8M\alpha)$, so for every i it contains a point $y_i \in S_{r_i\alpha}$, say $y_i \in D_{n_i}$. Choose $k_i \in K_i$ with $|y_i - \alpha k_i| < \alpha$, so that D_{n_i} meets $B(x_i, \alpha)$.

The events at $i < a$. The union $D_{n_i} \cup \cdots \cup D_n$ is connected, meets $B(x_i, \alpha)$ and contains a point outside $B(0, 8M\alpha)$, which contains $B(x_i, 8\alpha)$ because $r_i + 8 \leq 8M$ for $i < a$. Let $D_{i'}$, $i' \geq n_i$, be the first of these disks not contained in $B(x_i, 8\alpha)$. The centres of $D_{n_i}, \dots, D_{i'-1}$ lie in $B(x_i, 8\alpha)$, and the centre of $D_{i'}$ is within distance 1 of a point of $B(x_i, 8\alpha)$, because $D_{i'}$ meets $D_{i'-1}$ if $i' > n_i$ and meets $B(x_i, \alpha)$ if $i' = n_i$; so all these centres lie in R_i. Hence $B(x_i, \alpha) \cup D_{n_i} \cup \cdots \cup D_{i'}$ is a connected subset of $\Xi(R_i) \cup B(x_i, \alpha)$ containing x_i and not contained in $B(x_i, 8\alpha)$: this is $\mathcal{E}(x_i, \alpha)$.

The event at $i = a$. The union $D_1 \cup \cdots \cup D_{n_a}$ is connected, meets $B(x_a, \alpha)$ and contains a point of $B(0, M\alpha)$, which lies outside $B(x_a, 8\alpha)$ since $|x_a| = 8M\alpha$ and $7M > 8$. Running the chain backwards from D_{n_a}, the same argument gives $\mathcal{E}(x_a, \alpha)$. □

For $a \geq 3$ the intermediate circles could be used in either direction; we have used the forward direction for all $i < a$.

3 Decoupling through the shared lines

For pairwise disjoint bounded Borel sets $A_1, \dots, A_a$ let $\mathcal{S} = \mathcal{S}(A_1, \dots, A_a)$ be the set of lines of $\mathcal{L}$ meeting at least two of them, and $\mathcal{P}_i$ the set of lines of $\mathcal{L}$ meeting A_i and no other A_j (the *private* lines of A_i). For a finite set H of lines write $H \cap A := \{\ell \in H : \ell \cap A \neq \varnothing\}$.

Lemma 3.1 (Two levels of independence)**.** *Let $A_1, \dots, A_a$ be pairwise disjoint bounded Borel sets, G a finite set of lines, and let F_i be an event determined by the lines of $\mathcal{L} \cup G$ meeting A_i and the points of X in A_i. Then*

$$\mathbb{P}_G\Big(\bigcap_{i=1}^{a} F_i\Big) = \mathbb{E}\Big[\prod_{i=1}^{a} \mathbb{P}_{G \cup \mathcal{S}}(F_i)\Big]\,, \tag{3.1}$$

where inside the expectation $\mathcal{S}$ is treated as a deterministic set of fixed lines and $\mathbb{P}_{G\cup\mathcal{S}}(F_i)$ is the probability of F_i in the model whose random lines are the Poisson process $\mathcal{P}_i$ and whose fixed lines are $(G \cup \mathcal{S}) \cap A_i$.

Proof. The sets of lines meeting exactly one of the A_i, for each i, and the set of lines meeting at least two, are disjoint measurable sets of lines, so $\mathcal{P}_1, \dots, \mathcal{P}_a, \mathcal{S}$ are independent Poisson processes. Conditionally on all lines, the points of X in the disjoint sets $A_1, \dots, A_a$ are independent Poisson processes on the lines. Hence, conditionally on $\mathcal{S}$, the pairs $(\mathcal{P}_i, X \cap A_i)$, $i \leq a$, are independent, and F_i is determined by $(\mathcal{P}_i, X \cap A_i)$ together with the fixed lines $(G \cup \mathcal{S}) \cap A_i$. Taking the conditional expectation given $\mathcal{S}$ gives (3.1). □

Applied to the inclusion (2.2) with $A_i = R_i$,

$$\pi_{M\alpha}(x;G) \le \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\prod_{i=1}^{a} \pi_\alpha(x_i; G\cup\mathcal{S}_k)\Big], \qquad \mathcal{S}_k := \mathcal{S}(R_1,\dots,R_a)\,, \tag{3.2}$$

where only the lines of $G\cup\mathcal{S}_k$ meeting the respective region matter in each factor. This is the multi-region step. Its content is that the sub-crossings are independent given the shared lines, and that the shared lines enter as fixed lines of the sub-problems. In the Poisson Boolean model of [Gou08] the analogue of the shared lines is the set of balls of radius at least α, which is empty with high probability; here $\mathcal{S}_k$ is never empty, and its law is the object the recursion has to control.

Lemma 3.2 (Shared lines and aligned lines)**.** *Let $R = B(y,10\alpha)$ and $R' = B(y',10\alpha)$ with $|y-y'| \ge 70\alpha$, and $\alpha \ge 1$.*

(a) *The lines meeting R have ν-measure $20\mu\alpha$; a fortiori $\nu(\mathcal{S}(R,R')) \le 20\mu\alpha$.*

(b) *Let ℓ_0 be a line meeting R. The lines that pass within distance 1 of two points $p,q \in R$ with $|p-q| \ge \alpha$ of which ℓ_0 also passes within distance 1 have measure at most $\beta_1 := C_1\mu/\alpha$ with $C_1 := 368$.*

(c) *Let ℓ_0 be a line meeting R and R'. The lines that pass within distance 1 of a point $p \in R$ and a point $p' \in R'$ of which ℓ_0 also passes within distance 1 have measure at most $\beta_2 := C_2\mu/\alpha$ with $C_2 := 1/2$.*

Proof. (a) is the perimeter formula. (b) Let ℓ be such a line and φ its angle with ℓ_0. The points p,q are within distance 1 of both lines and at distance at least α, so $\alpha\sin\varphi \le 4$ and $\varphi \le (\pi/2)\sin\varphi \le 2\pi/\alpha$; the normal direction of ℓ lies in a set of measure at most $4\pi/\alpha$. Given the direction, ℓ passes within distance 2 of a point of the segment $\ell_0 \cap B(y,11\alpha)$, whose length is at most 22α, and along this segment the distance between ℓ and ℓ_0 varies by at most $22\alpha\sin\varphi \le 88$; so the offset of ℓ lies in an interval of length at most $4+88 = 92$. The measure is at most $(\mu/\pi)(4\pi/\alpha)\cdot 92 = 368\mu/\alpha$. (c) Now $|p-p'| \ge 50\alpha$, so $50\alpha\sin\varphi \le 4$ and $\varphi \le \pi/(25\alpha)$, the direction set has measure at most $2\pi/(25\alpha)$, the offset variation along $\ell_0 \cap B(y,11\alpha)$ is at most $22\alpha\cdot 4/(50\alpha) < 2$, the offset interval has length at most 6, and the measure is at most $(\mu/\pi)(2\pi/(25\alpha))\cdot 6 < \mu/(2\alpha)$. □

Any two of the regions $R_1,\dots,R_a$ have centres at distance at least 70α, so the lemma applies to every pair. The ratio $C_2/C_1 = 1/736$ measures how much rarer alignment across the gap between two regions is than alignment within one region; it plays the role that the radius of a ball serving both regions plays in [Gou08].

4 The recursion

4.1 Abstract form

Let $\alpha_1 \ge 1$. A *target* is a function $f\colon [\alpha_1,\infty) \to (0,1]$, and a *weight* is a measurable function $W_\alpha(x;G) \ge 1$ of $\alpha \ge \alpha_1$, $x \in \mathbb{R}^2$ and finite sets G of lines, depending on G only through $G \cap B(x,10\alpha)$.

Theorem 4.1 (Recursion)**.** *Let (f,W) be a target and a weight such that*

(B) (base case) $\pi_\alpha(x;G) \le f(\alpha)W_\alpha(x;G)$ *for all $\alpha \in [\alpha_1, M\alpha_1)$, $x \in \mathbb{R}^2$ and finite G;*

(R) (recursion) *for all* $\alpha \geq \alpha_1$, $x \in \mathbb{R}^2$ *and finite* G,

$$\sum_{k\in\mathcal{K}} \mathbb{E}\Big[\prod_{i=1}^{a} \min\{1, f(\alpha)W_\alpha(x_i; G\cup\mathcal{S}_k)\}\Big] \leq f(M\alpha)\, W_{M\alpha}(x; G)\,. \tag{4.1}$$

Then $\pi_\alpha(x;G) \leq f(\alpha)W_\alpha(x;G)$ *for all* $\alpha \geq \alpha_1$, $x \in \mathbb{R}^2$ *and finite* G. *In particular, if* $W_\alpha(x;\varnothing) \leq C_W$ *for all* α, x *and* $f(\alpha) \to 0$, *then* Ξ *has no unbounded component almost surely.*

Proof. Induction on n for $\alpha \in [M^n\alpha_1, M^{n+1}\alpha_1)$. The case $n = 0$ is (B). For $n \geq 1$ write $\alpha = M\alpha'$ with $\alpha' \in [M^{n-1}\alpha_1, M^n\alpha_1)$. By (3.2), the induction hypothesis applied inside the expectation with the fixed set $G\cup\mathcal{S}_k$, the trivial bound $\pi \leq 1$, and (R),

$$\pi_{M\alpha'}(x;G) \leq \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\prod_{i=1}^{a} \min\{1, f(\alpha')W_{\alpha'}(x_i; G\cup\mathcal{S}_k)\}\Big] \leq f(M\alpha')W_{M\alpha'}(x;G)\,.$$

The last assertion follows from Lemma 1.1. □

The theorem separates the probabilistic input from the arithmetic: (B) is a statement about bounded regions, and (R) is an inequality between deterministic functions in which the only random object is the Poisson process $\mathcal{S}_k$ of shared lines, whose law is explicit. In the Poisson Boolean model the analogue of (R) is $C\pi(\alpha)^2 + \lambda C\int_\alpha^\infty r^d \nu(\,\mathrm{d}r) \leq \pi(10\alpha)$, with no weight because the dependence is carried by an event of small probability.

4.2 The recursive weight, and the single remaining question

Nothing prevents us from *defining* f and W by (B) and (R).

Definition 4.2 (Recursive target and weight)**.** Fix $\alpha_1 \geq 1$. For $\alpha \in [\alpha_1, M\alpha_1)$ put $f(\alpha) := \pi_\alpha$ and $W_\alpha(x;G) := \max\{1, \pi_\alpha(x;G)/f(\alpha)\}$. For $\alpha \in [M^n\alpha_1, M^{n+1}\alpha_1)$, $n \geq 1$, with $\alpha' = \alpha/M$, put

$$f(\alpha) := \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\prod_{i=1}^{a} \min\{1, f(\alpha')W_{\alpha'}(\alpha' k_i; \mathcal{S}_k)\}\Big] \wedge 1\,, \tag{4.2}$$

$$W_\alpha(x;G) := \max\Big\{1,\ \frac{1}{f(\alpha)} \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\prod_{i=1}^{a} \min\{1, f(\alpha')W_{\alpha'}(x_i; G\cup\mathcal{S}_k)\}\Big]\Big\}\,, \tag{4.3}$$

where in (4.2) the regions are those of $x = 0$ at scale α'; by stationarity the value does not depend on x.

Proposition 4.3 (Unconditional crossing bound)**.** *With* f *and* W *of Definition 4.2,* $\pi_\alpha(x;G) \leq f(\alpha)W_\alpha(x;G)$ *for all* $\alpha \geq \alpha_1$, x *and* G, *and* $W_\alpha(x;\varnothing) = 1$ *whenever* $f(\alpha) < 1$*; in particular* $\pi_\alpha \leq f(\alpha)$. *Consequently* Ξ *has no unbounded component almost surely as soon as* $f(\alpha) \to 0$.

Proof. (B) holds by the definition at the base scales, (R) holds by (4.3), and for $G = \varnothing$ the maximum in (4.3) is attained by 1 because the sum equals $f(\alpha)$ by (4.2) when $f(\alpha) < 1$. Apply Theorem 4.1 and Lemma 1.1. □

Everything is now in one place.

Claim T (The recursive target tends to zero). There are $\mu_0 > 0$ and $\lambda_0 > 0$ such that for $\mu \le \mu_0$, $\lambda \le \lambda_0$, some $a \ge 2$ and a suitable α_1, the function f of Definition 4.2 satisfies $f(\alpha) \to 0$ as $\alpha \to \infty$.

Claim T is equivalent to the success of the scheme, and by Proposition 4.3 it implies $\lambda_c(\mu) \ge \lambda_0$. It is a statement about an explicit sequence of expectations over Poisson processes of lines: unrolling (4.2) and (4.3), $f(M^n\alpha_1)$ is an expectation over a tree of nested a-tuples of regions, the shared lines of each tuple being passed to the sub-regions as fixed lines, and at the leaves of the tree the base-scale probabilities $\pi_\alpha(x; G)$ with $\alpha \in [\alpha_1, M\alpha_1)$ and G the accumulated shared lines. We do not prove the claim. The rest of this note shows that one step of the recursion closes under a hypothesis on the weights of the shared lines (Section 5), and formulates the inductive statement that would prove the claim (Section 6).

Remark 4.4 (The order of the target). Since $f(\alpha) \ge \pi_\alpha$, the target cannot decay faster than the crossing probability, and the following computation shows how fast that is. A crossing of the annulus from $B(0,\alpha)$ to $\partial B(0, 8\alpha)$ by unit disks whose centres lie on lines requires, if all centres lie on m lines running together within distance 1 across the annulus, a sequence of points with consecutive gaps at most 2 along a length of order α in a Poisson process of intensity $m\lambda$, which has probability of order $\exp\{-c\alpha\, m\lambda \mathrm{e}^{-2m\lambda}\}$; the m lines themselves cost $(\mu/\alpha)^m\alpha^2$ up to constants. Optimising over m gives $2m\lambda \approx \log\alpha$ and $\pi_\alpha \approx \exp\{-(\log\alpha)^2/(2\lambda)\}$ up to lower-order terms: faster than any power of α, and with no fixed number j_0 of lines. The recursion will produce a target of this type: the error term of one step is dominated by bundles of shared lines, so that $f(M\alpha)$ is of the order of $M^{2-m^*(\alpha)}f(\alpha)$ with $m^*(\alpha)$ of order $\log\alpha$ (Section 5), which integrates to $\exp\{-c(\log\alpha)^2\}$.

Remark 4.5 (A constant number of regions suffices: a correction). Our multiscale note asserted that a polynomial target forces a recursion with many well-separated sub-boxes at scales $L \mapsto L^3$, because with two sub-boxes the exponent in the basic inequality is negative. That assertion is wrong. The exponent computed there contained the positional entropy $(L/\ell)^{2a}$ of the sub-boxes, which is polynomial in ℓ when $L = \ell^3$ but is the constant $N_{\mathcal{K}}$ at a fixed ratio; at a fixed ratio the product term $f(\alpha)^a$ is below any polynomial or quasi-polynomial target $f(M\alpha)$ by a factor tending to zero, and the recursion is driven by the error term, exactly as in [Gou08]. The large number of sub-boxes was also what made the nearly complete groups of that note hard: along a row of fixed lines the sub-boxes have positional entropy $(L/\ell)^a$, which the gain of a group with few missing lines could not pay. With a constant number of regions the corresponding term is $f(\alpha)^2 k_\alpha(m)^2$ against $f(M\alpha)k_{M\alpha}(m)$, and it closes for every group size (Section 6).

5 One recursion step under a group-weight hypothesis

In this section $\alpha \ge \alpha_1$ is fixed, $x = 0$, and $f = f(\alpha)$. We show that the step from α to $M\alpha$ for the target, i.e. the sum in (4.2), is at most $f/2$ plus an explicit *across-alignment term*, provided the weights of the shared-line configurations at scale α have the form one expects: groups of aligned lines are charged by their gain, with the cap 1. Since a product of a factors in $[0,1]$ is at most the product of any two of them, we bound each summand of (4.2) by the product of the factors for the regions R_1 and R_a; the other $a-2$ factors are only used in Section 6.

Bundles and group weights. For a region $R = B(y, 10\alpha)$ and a finite set H of lines, an *m-bundle of H in R* is a set of m lines of H that all pass within distance 1 of two points $p, q \in R$ with $|p - q| \geq \alpha$. Let $N_m(y; H)$ be the number of m-bundles of H in R; $N_1(y; H)$ is the number of lines of H meeting R. For $A > 0$ put

$$k_\alpha(m) := \frac{1}{\alpha^2}\Big(\frac{A\alpha}{\mu}\Big)^m, \qquad \kappa_m := \min\{1, f\, k_\alpha(m)\}\,. \tag{5.1}$$

This is the gain one expects from m fixed aligned lines when a crossing is carried by bundles: each fixed line replaces a random line, which by Lemma 3.2(b) costs a factor of order μ/α, so that A is of the order of $1/C_1$, and the bundle loses its freedom of position, a factor α^2.

Definition 5.1 (Group-weight hypothesis at scale α)**.** $(\mathrm{GW})_\alpha$ holds with parameters $m^* \geq 2$ and $A > 0$ if for every $y \in \mathbb{R}^2$ and every finite set H of lines,

$$\min\{1, f\, W_\alpha(y; H)\} \leq f + \sum_{m=1}^{m^*} \kappa_m\, N_m(y; H) \qquad \text{with } \kappa_{m^*} := 1\,. \tag{5.2}$$

The hypothesis says that the weight of a configuration is at most the group weight, groups of m^* or more lines being charged nothing beyond the cap. Put

$$p_m := \frac{20\mu\alpha\, \beta_1^{m-1}}{m!}, \qquad q_{m,m',j} := 20\mu\alpha\, \beta_2^{j-1}\beta_1^{m+m'-2j} \quad (1 \leq j \leq m \wedge m')\,. \tag{5.3}$$

Lemma 5.2 (Bundle counts)**.** *Let $k \in \mathcal{K}$, $\mathcal{S} = \mathcal{S}_k$, $N_m := N_m(\alpha k_1; \mathcal{S})$ and $N'_{m'} := N_{m'}(\alpha k_a; \mathcal{S})$, the bundle counts in R_1 and R_a. Then $\mathbb{E}N_m \leq p_m$, and*

$$\mathbb{E}[N_m N'_{m'}] \leq p_m p_{m'} + \sum_{j=1}^{m\wedge m'} q_{m,m',j}\,. \tag{5.4}$$

Proof. An m-bundle in R_1 is an m-subset of $\mathcal{S}$ whose lines meet R_1. Ordering it, the first line meets R_1, measure at most $20\mu\alpha$, and each further line passes within distance 1 of two points of R_1 at distance at least α of which the first line also passes within distance 1, measure at most β_1 by Lemma 3.2(b); the multivariate Mecke formula [LP17, Theorem 4.4] gives $\mathbb{E}N_m \leq 20\mu\alpha\beta_1^{m-1}/m!$. For the pairs (T, T'), split according to $j = |T \cap T'|$. The pairs with $j = 0$ are pairs of disjoint subsets, and the Mecke formula for disjoint tuples gives exactly the product of the two integrals, at most $p_m p_{m'}$. For $j \geq 1$, order the configuration so that a common line comes first, measure at most $20\mu\alpha$; each of the other $j - 1$ common lines passes within distance 1 of a point of R_1 and of a point of R_a of which the first line also passes within distance 1, because both belong to bundles in both regions, measure at most β_2 by Lemma 3.2(c); and each of the $m - j$ remaining lines of T and $m' - j$ remaining lines of T' has measure at most β_1. The number of pairs is at most the number of such ordered configurations. □

Proposition 5.3 (One step)**.** *Assume $(\mathrm{GW})_\alpha$ with parameters m^* and A. Then*

$$\begin{aligned}\sum_{k\in\mathcal{K}} \mathbb{E}\Big[\prod_{i=1}^{a} \min\{1, fW_\alpha(\alpha k_i; \mathcal{S}_k)\}\Big] &\leq \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\min\{1, fW_\alpha(\alpha k_1; \mathcal{S}_k)\}\min\{1, fW_\alpha(\alpha k_a; \mathcal{S}_k)\}\Big] \\ &\leq N_{\mathcal{K}}\Big[(f + \Pi)^2 + Q_1 + \Lambda\Big]\,,\end{aligned} \tag{5.5}$$

where, with $\sigma_j := \sum_{m=j}^{m^*} \kappa_m \beta_1^{m-j}$,

$$\Pi := \sum_{m=1}^{m^*} \kappa_m p_m, \qquad Q_1 := 20\mu\alpha\,\sigma_1^2, \qquad \Lambda := \sum_{j=2}^{m^*} 20\mu\alpha\,\beta_2^{j-1}\sigma_j^2\,. \tag{5.6}$$

Proof. Fix $k \in \mathcal{K}$ and use the notation of Lemma 5.2. By (5.2) and $\mathbb{E}N_m \le p_m$,

$$\begin{aligned}\mathbb{E}\Big[\Big(f + \sum_m \kappa_m N_m\Big)\Big(f + \sum_{m'} \kappa_{m'} N'_{m'}\Big)\Big] &\le f^2 + 2f\Pi + \sum_{m,m'} \kappa_m\kappa_{m'}\mathbb{E}[N_m N'_{m'}] \\ &\le (f+\Pi)^2 + \sum_{m,m'} \kappa_m\kappa_{m'} \sum_{j\ge 1} q_{m,m',j}\end{aligned}$$

by (5.4). For fixed j, $\sum_{m,m'\ge j} \kappa_m\kappa_{m'} q_{m,m',j} = 20\mu\alpha\beta_2^{j-1}\big(\sum_{m\ge j}\kappa_m\beta_1^{m-j}\big)^2 = 20\mu\alpha\beta_2^{j-1}\sigma_j^2$, which is Q_1 for $j=1$ and the j-th term of Λ for $j \ge 2$. Sum over $k \in \mathcal{K}$. □

Lemma 5.4 (Size of the terms)**.** *Let* $C_A := (20/C_1)(\mathrm{e}^{AC_1} - 1)$. *Then*

$$\Pi \le fC_A + p_{m^*}, \qquad Q_1 \le \frac{40\,\mathrm{e}^{2AC_1}}{C_1^2\,\mu\alpha}\,f^2 + \frac{(m^*!)^2}{10\,\mu\alpha}\,p_{m^*}^2\,. \tag{5.7}$$

Moreover, if $\beta_1 \le 1/2$ *and* $m_0 := \min\{m\colon fk_\alpha(m) \ge 1\} \wedge m^*$, *then*

$$\sigma_j \le m_0\, f\,\mu^{-j}\alpha^{j-2} \max_{j\le m<m_0} A^m C_1^{m-j} + 2\beta_1^{m_0-j} \quad (j \le m_0), \qquad \sigma_j \le 2 \quad (m_0 < j \le m^*)\,.$$

Proof. $\kappa_m p_m \le fk_\alpha(m)p_m = f\cdot 20A^m C_1^{m-1}/m!$ for $m < m^*$, and $\kappa_{m^*}p_{m^*} = p_{m^*}$; summing gives the bound on Π. For Q_1, $\sigma_1 \le \sum_{m<m^*} fk_\alpha(m)\beta_1^{m-1} + \beta_1^{m^*-1} = f\sum_{m<m^*} A^m C_1^{m-1}/(\mu\alpha) + \beta_1^{m^*-1} \le f\mathrm{e}^{AC_1}/(C_1\mu\alpha) + p_{m^*}m^*!/(20\mu\alpha)$, and $(a+b)^2 \le 2a^2+2b^2$. For σ_j, split the sum at m_0: for $m < m_0$, $\kappa_m\beta_1^{m-j} \le fk_\alpha(m)\beta_1^{m-j} = fA^m C_1^{m-j}\mu^{-j}\alpha^{j-2}$, and there are fewer than m_0 such terms; for $m \ge m_0$, $\kappa_m \le 1$ and the geometric sum is at most $\beta_1^{(m_0-j)^+}/(1-\beta_1) \le 2\beta_1^{(m_0-j)^+}$. □

Corollary 5.5 (Halving)**.** *Assume* $(\mathrm{GW})_\alpha$, $\beta_1 \le 1/2$, *and*

$$p_{m^*} \le f, \qquad f \le \frac{1}{16N_{\mathcal{K}}(2+C_A)^2}, \qquad Q_1 \le \frac{f}{8N_{\mathcal{K}}}, \qquad \Lambda \le \frac{f}{8N_{\mathcal{K}}}\,. \tag{5.8}$$

Then the sum in (5.5) *is at most* $f/2$, *so that* $f(M\alpha) \le f(\alpha)/2$ *in Definition 4.2.*

Proof. $(f+\Pi)^2 \le (f(1+C_A)+p_{m^*})^2 \le f^2(2+C_A)^2 \le f/(16N_{\mathcal{K}})$; add the bounds on Q_1 and Λ and multiply by $N_{\mathcal{K}}$. □

Reading the conditions. The first condition in (5.8) says that bundles large enough to be treated as free are rarer than the target; it defines the natural choice

$$m^* := \min\{m \ge 2\colon p_m \le f\}, \qquad \text{for which } p_{m^*} \le f < p_{m^*-1} \text{ and } m^* \approx \frac{\log(1/f)}{\log(\alpha/\mu)}\,. \tag{5.9}$$

The second is smallness of the target. The third holds for $\alpha \ge \alpha_0(\mu, A)$ once $p_{m^*} \le f$, because by (5.7) $Q_1 \le f\cdot[40\mathrm{e}^{2AC_1}f/(C_1^2\mu\alpha) + (m^*!)^2 f/(10\mu\alpha)]$ and $f < p_{m^*-1} \le 20\mu\alpha\beta_1^{m^*-2} = 20C_1^{m^*-2}\mu^{m^*-1}\alpha^{3-m^*}$ makes the bracket at most $C^{m^*}\alpha^{2-m^*}$, small for $m^* \ge 3$ and α large. The fourth condition is the substantial one. The dominant contributions to Λ come from the indices

$j \geq m_0$, where the common lines are charged with $\kappa = 1$: by Lemma 5.4, $\sum_{j \geq m_0} 20\mu\alpha\beta_2^{j-1}\sigma_j^2 \leq 160\mu\alpha\beta_2^{m_0-1}$, which is the probability that m_0 shared lines are aligned across both regions, up to the factor 8. The condition $\Lambda \leq f/(8N_{\mathcal{K}})$ therefore requires

$$160\,\mu\alpha\Big(\frac{C_2\mu}{\alpha}\Big)^{m_0-1} \leq \frac{f}{16N_{\mathcal{K}}}\,, \tag{5.10}$$

while m_0 is defined by $fk_\alpha(m_0) \geq 1$, i.e. $f \geq \alpha^2(\mu/(A\alpha))^{m_0}$. Both hold together if $\alpha^2(\mu/(A\alpha))^{m_0} \geq 2560N_{\mathcal{K}}\mu\alpha(C_2\mu/\alpha)^{m_0-1}$, that is, if $(AC_2)^{-m_0} \geq 2560N_{\mathcal{K}}/C_2$, which is true for all $m_0 \geq m_1(A)$ provided $AC_2 < 1$. With A of the order of $1/C_1$ this is comfortably the case, and the terms $j < m_0$ of Λ are handled by the first bound of Lemma 5.4 in the same way as Q_1. In words: a configuration of m_0 shared lines aligned across both regions makes both sub-crossings free and has probability $20\mu\alpha\beta_2^{m_0-1}$; the scheme can only work if the target at scale α is already at least this large, which it is once the gain constant A is below $1/C_2$, because then the same m_0 lines aligned within a single region, which cost β_1 per line instead of β_2, are a configuration the target must already account for. We have not verified $(\mathrm{GW})_\alpha$ itself; see the next section.

6 What has to propagate

Proposition 5.3 concerns one step and assumes $(\mathrm{GW})_\alpha$ for the recursively defined weights of the configurations $\mathcal{S}_k$. The scheme succeeds if a hypothesis of this kind propagates from scale α to scale $M\alpha$. We first record what the recursion (4.3) assigns to the basic kinds of fixed-line configurations, writing $f = f(\alpha)$ and using $f(M\alpha) \approx M^{2-m^*}f$ from Section 5 as the order of the target at the next scale, and $f < p_{m^*-1} \leq 20C_1^{m^*-2}\mu^{m^*-1}\alpha^{3-m^*}$ from (5.9). We then formulate the inductive statement.

A group serving one region. Let G be an m-bundle in R_1 and in no other region, $m < m^*$. Under $(\mathrm{GW})_\alpha$ the tuple contributes to (4.3) at most $fk_\alpha(m)\cdot(f(1+C_A))^{a-1}$ up to the shared-line corrections of Section 5, against the target's $f(M\alpha)k_{M\alpha}(m)$. The ratio is $f^{a-1}(1+C_A)^{a-1}M^{m^*-2}k_\alpha(m)/k_{M\alpha}(m) = f^{a-1}(1+C_A)^{a-1}M^{m^*-m}$, which is small already for $a = 2$.

A group serving all regions. Let G be an m-bundle in every R_i, i.e. aligned across the whole span $7M\alpha$. The contribution is $(fk_\alpha(m))^a$ against $f(M\alpha)k_{M\alpha}(m)$, ratio

$$\begin{aligned} f^{a-1}M^{m^*-2}\frac{k_\alpha(m)^a}{k_{M\alpha}(m)} &= f^{a-1}M^{m^*-m}\Big(\frac{A\alpha}{\mu}\Big)^{(a-1)m}\alpha^{-2(a-1)} \\ &\leq \Big[20C_1^{m^*-2}A^m M^{\frac{m^*-m}{a-1}}\mu^{m^*-1-m}\alpha^{1+m-m^*}\Big]^{a-1}\,, \end{aligned}$$

small for $m \leq m^* - 2$ and bounded for $m = m^* - 1$. This is the case that in our multiscale note required many sub-boxes and a separate lemma for nearly complete groups; here it closes because the positional entropy is a constant (Remark 4.5).

A group making r regions free. Let G contain m^*-bundles at scale α in r of the regions and nothing else of note. The contribution is at most $(f(1+C_A))^{a-r}$. If $r \leq a-2$, this is at most $f^2(1+C_A)^2$, and the ratio to $f(M\alpha)$ is at most $f(1+C_A)^2M^{m^*-2} \leq 20(1+C_A)^2C_1^{m^*-2}M^{m^*-2}\mu^{m^*-1}\alpha^{3-m^*}$, small for $\alpha \geq \alpha_0(M)$: *no weight is needed*, the recursive weight of such G is 1. If $r \geq a-1$, the contribution is at least $f(1+C_A)$ or 1, and the recursion assigns

$W_{M\alpha}(x;G) \geq M^{m^*-2}(1+C_A)/N_{\mathcal{K}}$. This is far more than the true gain of such a configuration, which is only of order $(7M/(7M-2r))^{m^*}$, because the free stretches have length of order α out of a span $7M\alpha$; and it is also more than the group weight at scale $M\alpha$, since an m^*-bundle at scale α is not a bundle at scale $M\alpha$ at all, its alignment length being too short. So $(\mathrm{GW})_{M\alpha}$ fails for these configurations: the pointwise group-weight hypothesis does not propagate.

Why this is harmless for $a \geq 3$. The recursion does not need (GW) for every configuration, only in expectation over the shared lines of the level above. At scale $M\alpha$ the fixed set of a sub-region is $G \cup \mathcal{S}_{k'}$ with $\mathcal{S}_{k'}$ the shared lines of a tuple of regions of radius $10M\alpha$. For $\mathcal{S}_{k'}$ to make $a-1$ of the a sub-regions of radius 10α inside one of its regions free, it must contain $m^*(\alpha)$-bundles at scale α in $a-1$ of them; by Lemma 5.2 applied at scale α and the Mecke formula for disjoint tuples, the probability of this is at most $(N_{\mathcal{K}} p_{m^*})^{a-1} \leq (N_{\mathcal{K}} f)^{a-1}$ up to the overlap terms of Section 5. Since $\min\{1,\cdot\} \leq 1$, such configurations contribute at most their probability to the expectation at scale $M^2\alpha$, which is at most $(N_{\mathcal{K}} f(\alpha))^{a-1}$, to be compared with the target $f(M^2\alpha) \approx M^{4-m^*(\alpha)-m^*(M\alpha)} f(\alpha)$: for $a \geq 3$ the ratio is at most $N_{\mathcal{K}}^{a-1} f(\alpha) M^{m^*(\alpha)+m^*(M\alpha)}$, which is small by the bound on f. For $a = 2$ the ratio is of order M^{m^*}, and the tile configurations cannot be discarded. This is the reason for the multi-region inclusion of Section 2.

The inductive statement. The hypothesis that should propagate is therefore the group-weight bound in expectation, with an exceptional set. For a finite set G of lines, a tuple $k \in \mathcal{K}$ of regions at scale α with centres x_i, and the shared lines $\mathcal{S}_k$ of the tuple, say that $(\mathrm{GW})^{\mathrm{E}}_{\alpha}(G,k)$ holds if

$$\mathbb{E}\Big[\prod_{i=1}^{a} \min\{1, fW_\alpha(x_i; G \cup \mathcal{S}_k)\}\Big] \leq \mathbb{E}\Big[\prod_{i=1}^{a}\Big(f + \sum_{m=1}^{m^*} \kappa_m N_m(x_i; G \cup \mathcal{S}_k)\Big)\Big] + \varepsilon_\alpha(G)\,, \tag{6.1}$$

with an error $\varepsilon_\alpha(G)$ that is at most $f(M\alpha)/(4N_{\mathcal{K}})$ for $G = \varnothing$ and, for general G, at most a quarter of what the right-hand side of (4.3) has to produce for G. For $G = \varnothing$, (6.1) is exactly what the proof of Proposition 5.3 uses, so that $(\mathrm{GW})^{\mathrm{E}}_{\alpha}(\varnothing, k)$ for all k and the conditions of Corollary 5.5 give $f(M\alpha) \leq f(\alpha)/2 + \dots$ as there. Claim T would follow from the propagation

$$\big[(\mathrm{GW})^{\mathrm{E}}_{\alpha}(G,k) \;\; \forall G \in \mathcal{G}_\alpha,\; \forall k\big] \implies \big[(\mathrm{GW})^{\mathrm{E}}_{M\alpha}(G',k') \;\; \forall G' \in \mathcal{G}_{M\alpha},\; \forall k'\big]\,, \tag{6.2}$$

together with its validity at the base scale, where the class $\mathcal{G}_\alpha$ has to contain the empty set and the sets $G' \cup \mathcal{S}_{k'}$ that occur as fixed sets of sub-regions, up to events of probability absorbed in ε. The computations above are the three cases of this implication for the simplest configurations; what is missing is the general case, in which G' is an arbitrary finite set and the bundles of $G' \cup \mathcal{S}_{k'}$ in the sub-regions have to be related to the bundles of G' in the big region and to the shared lines. The structure of the argument is the one of Proposition 5.3: expand the product, use the Mecke formula on ordered tuples of lines, and separate the lines of G', which are deterministic, from those of $\mathcal{S}_{k'}$, which are Poisson with the explicit intensity of Lemma 3.2. This is the next task.

The points. The Cox points enter only through the base case and through the caps $\min\{1,\cdot\}$: the recursion never asks how likely it is that a bundle of lines actually carries occupied disks, and a bundle of $m^*(\alpha)$ lines is treated as free. This is a loss when $m^*\lambda$ is small, since then the Cox process on the bundle has gaps larger than 2 at rate $m^*\lambda \mathrm{e}^{-2m^*\lambda}$ per unit length and the crossing of length 7α along the bundle is unlikely. The loss is affordable, because the recursion only needs $f \to 0$ and the bundle probabilities p_{m^*} already decay like $(\mu/\alpha)^{m^*}$ with $m^* \to \infty$; the parameter λ then enters only through the base case at scale α_1, where it must make π_{α_1} small.

A sharper scheme would carry the point intensity on a bundle in the weight, at the price of a two-parameter weight.

Relation to the referee document. The present scheme supersedes the skeleton bound, the first moment over cluster chains and the dependence step of the referee document: there is no thick cluster and no relay, since a bundle of lines released by one region is just a set of fixed lines for the next, charged by its weight. The multiscale lemmas of that document survive in simplified form, with a constant number of regions at a fixed ratio in place of a sub-boxes at ratio $L^{2/3}$, and the coverage lemma and the base case remain the tools for the two statements still open here: the base case (B) at scale α_1, and the propagation (6.2), which is Claim T.

7 Summary

1. Gouéré's inclusion holds for the Cox Boolean model without an error event and with any constant number $a \geq 2$ of bounded regions of radius 10α on circles 70α apart, at scale ratio $M = 10(a-1)$ (Lemma 2.1); the sub-crossings are independent given the lines meeting at least two regions, which become fixed lines of the sub-problems (Lemma 3.1).

2. The recursion is a deterministic inequality (R) for a target f and a weight W, in which randomness enters only through the Poisson process of shared lines (Theorem 4.1). Defining f and W recursively from the base case makes the bound $\pi_\alpha \leq f(\alpha)$ unconditional (Proposition 4.3), and the subcritical phase follows from the single statement $f(\alpha) \to 0$ (Claim T).

3. One step of the recursion closes, $f(M\alpha) \leq f(\alpha)/2$, if the weights of shared-line configurations at scale α are bounded by group weights with bundles of size $m^*(\alpha)$ treated as free (Proposition 5.3, Corollary 5.5); the configurations aligned across two regions are controlled by the factor C_2/C_1 between alignment across the gap and alignment within a region, and the substantial condition is $AC_2 < 1$ for the gain constant.

4. Group weights propagate for groups serving one or all regions, which removes the nearly complete groups and the many sub-boxes of the multiscale note (Remark 4.5), and configurations making at most $a-2$ regions free need no weight at all. Configurations making $a-1$ regions free violate the pointwise group-weight bound at the next scale, but for $a \geq 3$ they occur among shared lines with probability of order $f(\alpha)^{a-1}$, far below the target, so that the group-weight bound in expectation, (6.1), can propagate. Proving the propagation (6.2) and the base case is what remains.

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Use of AI tools

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